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Oscillations in a Simple Microvascular Network

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Abstract—We have identified the simplest topology that will permit spontaneous oscillations in a model of microvascular blood flow that includes the plasma skimming effect and the Fahraeus–Lindqvist effect and assumes that the flow can be described by a first-order wave equation in blood hematocrit. Our analysis is based on transforming the governing partial differential equations into delay differential equations and analyzing the associated linear stability problem. In doing so we have discovered three dimensionless parameters, which can be used to predict the occurrence of nonlinear oscillations. Two of these parameters are related to the response of the hydraulic resistances in the branches to perturbations. The other parameter is related to the amount of time necessary for the blood to pass through each of the branches. The simple topology used in this study is much simpler than networks found *in vivo*. However, we believe our analysis will form the basis for understanding more complex networks.

Keywords—Oscillations, Microvascular network, Bloodflow.

INTRODUCTION

Temporal fluctuations are often observed in the microcirculation. Oscillations in red cell velocity[14] and pressure[24] have been observed in a variety of tissues[3,4,13,19] in many different animal models. Oscillatory flow has also been measured in humans.[22] Fluctuations in pressure and velocity have been shown to affect vascular resistance, fluid filtration rates[19] and mass transfer between tissue and the blood stream.[18]

Observed oscillation frequencies range up to 240 cycles per minute. High frequency oscillations (greater than 50 cycles per minute) have been attributed to either heart pulse or breathing rhythms. Lower frequency oscillations are thought to be caused by vasomotion.[19,22] Observed frequencies of vasomotion have been reported to range from 2.7 to 32 cycles per minute.[6,23] Recent analysis of RBC velocities and arteriole diameter dynamics by Parthimos *et al.*[19] suggests, however, that low frequency oscillations may not be solely due to vasomotion. Their analysis demonstrates that two important measures (correlation dimension and Lyapunov exponent) of the RBC velocity oscillations depends on whether vasomotion is present or not. This indicates that something other than vasomotion is also driving the RBC velocity dynamics.

About 10 years ago, Kiani *et al.*[16] combined information about plasma skimming,[17] the Fahraeus effect,[8] the Fahraeus–Lindqvist effect,[9] and network structure into a numerical simulation study of microvascular network function. Their computations for large, real network geometries indicated that spontaneous temporal oscillations of blood flow parameters (flow rate, pressure, hematocrit) are possible in microvascular networks. These oscillations, which occurred in approximately 30% of the vessels, are independent of any biological control. Despite significant numerical experimentation, they were unable to identify the parameters that could be used to predict this unusual dynamic behavior.

Carr and LeCoin[5] reformulated the same problem into a set of coupled partial differential equations (PDEs). Rather than considering networks of hundreds of vessel segments as Kiani *et al.* did, Carr and LeCoin studied smaller networks with 8–15 vessel segments. The results of Carr and LeCoin corroborated those of Kiani *et al.* Fixed points, damped oscillations and sustained oscillations were all evident in the simulations. Unfortunately, Carr and LeCoin were also unsuccessful in their attempt to discover the parameters that determine the dynamics of these networks. However, they did report some conditions which are necessary for oscillations to occur. First, only arcade type networks (not tree networks) permit oscillations. Second, both the Fahraeus–Lindqvist effect and plasma skimming are necessary for oscillations. Third, oscillations require large Peclet numbers (axial dispersion of red cells must be small compared to axial convection). Fourth, the frequency of the oscillation is related to the residence times of the blood in the vessel segments (the nature of that relationship was not known.) Fifth, the form of the viscosity function describing the Fahraeus–Lindqvist effect is not important. A linear approximation to the viscosity function near the steady

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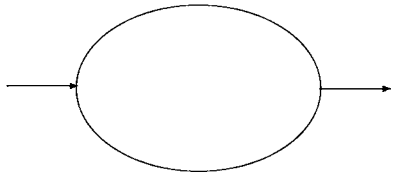


FIGURE 1. This network with only two interior nodes is the simplest network that can exhibit spontaneous oscillations. Only network topology is shown here—the actual network geometry is not implied.

state produces almost identical results as more complicated viscosity functions. Sixth, oscillations only occur when no steady state exists. As we shall see this last claim is erroneous. In addition to these necessary conditions, Carr and LeCoin showed that the Fahraeus effect is not necessary for oscillations to occur. Including the Fahraeus effect altered the frequency of the oscillations but did not cause them to disappear.

We have studied the simplest possible arcade network shown in Fig. 1. Our objective for this study is to gain insight into the important parameters governing spontaneous oscillations in microvascular networks. It consists of a ring with one inlet and one outlet branch. We have been able to transform the coupled PDEs for this network problem into a state dependent delay differential equation (DDE). By analyzing this DDE we have been able to identify the controlling dimensionless parameters that determine whether or not spontaneous oscillations will occur and to estimate their frequencies. We have identified two conditions with regard to the controlling parameters which are necessary for oscillations to occur. Some physical interpretations for these parameters and conditions are also offered.

MODELING PRELIMINARIES AND STEADY-STATE SOLUTIONS

The computational scheme of Kiani *et al.*,[16] as well as that of Carr and LeCoin,[5] first solves for the nodal pressures in the network, then the flow rate in each of the branches is found and finally the red cell convection in the tubes is determined. In the approach used here all of these computations will be done simultaneously. As a result, different model equations arise for different network topologies. All discussion in this paper concerns the simple topology shown in Fig. 1. The solution of the total mass balance and momentum balance for this network is

$$Q(t) = \frac{R_{\mathrm{B}}(t)}{R_{\mathrm{A}}(t) + R_{\mathrm{B}}(t)},$$

where $Q(t) = Q_{\mathrm{A}}(t)/Q_1$ is the fractional volumetric flow into daughter branch A at time t, $Q_{\mathrm{A}}(t)$ is the actual volumetric flow into branch A, Q_1 is the volumetric flow in the inlet, R_{A} is the hydraulic resistance in branch A, and R_{B} is the hydraulic resistance in branch B. In general, the resistance in a vessel is

$$R_i(t) = \int_0^{l_i} \frac{128\ \mu(H_i(x_i, t), d_i)}{\pi d_i^4}\, dx_i\ ,$$

where μ_i is the viscosity in branch i, $H_i(x_i, t)$ the hematocrit in branch i at location x_i and time t, l_i the length of branch i, and d_i the diameter of branch i. This integral accounts for the variation in hematocrit over the length of the tube. The mathematics are simplified if the viscosity is taken as a function of the axially averaged hematocrit,

$$\bar{H}_i(t) = \frac{1}{l_i}\int_0^{l_i} H_i(x_i, t)\, dx_i\ .$$

We have made this simplification in this study, but we have also conducted simulations which demonstrate that the conditions for oscillations are the same whether the hematocrit is averaged or the resistance is integrated along the length of the vessel. With this simplification the resistance in branch i becomes

$$R_i(t) = \frac{128\ l_i\ \mu_i(\bar{H}_i(t))}{\pi d_i^4}. \tag{1}$$

The Fahraeus–Lindqvist effect indicates that the apparent viscosity of blood in the microcirculation is a function of the hematocrit in the vessel and the vessel diameter. Several models for this viscosity effect are available in the literature.[1,2,15,21] In this study, any model used for the Fahraeus–Lindqvist effect can lead to oscillatory dynamics if it has the following properties: the viscosity, μ, must always be positive; the derivative of viscosity with respect to hematocrit, $\beta = d\mu/d\bar{H}$, must also be positive.

The plasma skimming rule determines the hematocrit of the blood at the entrance of the daughter branches of a diverging bifurcation. If we arbitrarily label one of the daughter branches A and the other B then the entrance hematocrit in branches A and B depend on the feed hematocrit and the volumetric flow into that branch. We may express the plasma skimming rule in the form,

$$H_{\mathrm{A}}(0, t) = H_{\mathrm{f}} f(Q(t)), \tag{2}$$

$$H_{\mathrm{B}}(0, t) = H_{\mathrm{f}} g(Q(t)), \tag{3}$$

where $Q(t)$ is the fractional volumetric flow into branch A, $H_{\mathrm{A}}(0, t)$ is the entrance hematocrit into branch A, H_{f} is the feed hematocrit, and f is the plasma skimming function for branch A. Conservation of mass dictates that $g(Q(t)) = f(1 - Q(t))$. This formulation allows us to use any of the several plasma skimming models available in the literature.[7,10,11,20] An example is the logit function suggested by Dellimore *et. al.*[7]

$$f(Q) = \frac{Q^{p-1}}{Q^p + (1-Q)^p}\ , \tag{4}$$

$$g(Q) = \frac{(1-Q)^{p-1}}{Q^p + (1-Q)^p}\ , \tag{5}$$

where p is a plasma skimming parameter. Notice that all plasma skimming functions depend on $Q(t)$. We will show later that $\bar{H}_{\rm A}$ and $\bar{H}_{\rm B}$ are determined from the history of the entrance hematocrit to each branch, $H_{\rm A}(0, t)$ and $H_{\rm B}(0, t)$, which implies, via the plasma skimming rule, that the average hematocrit in branches A and B is a function of the history of $Q(t)$. This allows us to rewrite the combined total mass and momentum balance as

$$Q(t) = \frac{R_{\rm B}(Q(t))}{R_{\rm A}(Q(t)) + R_{\rm B}(Q(t))} \ . \tag{6}$$

The steady-state flow rate, Q^*, can be determined by solving Eq. (6) subject to Eqs. (1)–(3). To compute the steady-state solution(s) the network parameters, $d_{\rm A}$, $d_{\rm B}$, $l_{\rm A}$, $l_{\rm B}$, and $H_{\rm f}$ are first specified along with the plasma skimming model and the viscosity model. Steady states can be demonstrated graphically by plotting both the LHS and the RHS of Eq. (6) against assumed values for Q^* on the same graph. The value of the abscissa where the two curves cross is the steady-state value of Q^*. An example is shown in Fig. 2. In general, a steady-state solution always exists for the following reasons. The RHS of Eq. (6) is a continuous function of Q. If we assume that the viscosity is always positive for any value of Q, then the RHS takes values between 0 and 1 for all values of Q (since flow resistance is never zero the RHS never takes the values of 0 or 1). Continuity of the RHS implies that at least one steady-state solution exists. The speculation by Carr and LeCoin that oscillations occur only if steady states do not exist is therefore erroneous.

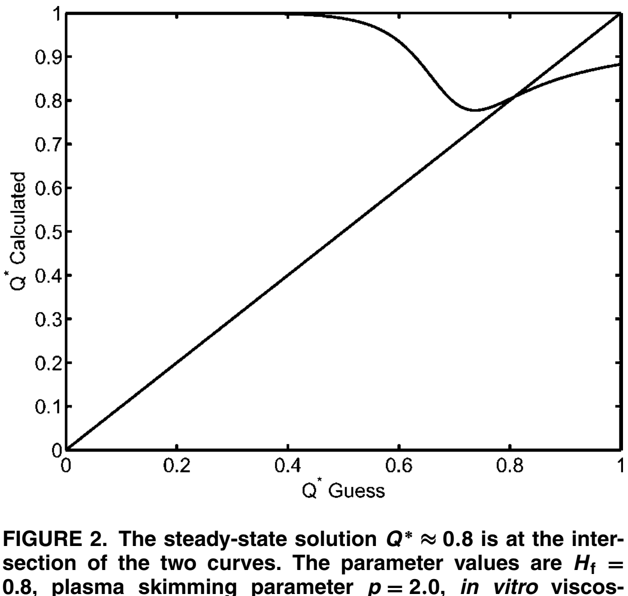


FIGURE 2. The steady-state solution $Q^* \approx 0.8$ is at the intersection of the two curves. The parameter values are $H_{\rm f} = 0.8$, plasma skimming parameter $p = 2.0$, *in vitro* viscosity model,[21] $d_{\rm A} = 35\,\mu$m, $d_{\rm B} = 20\,\mu$m, $l_{\rm A} = 50\,\mu$m, and $l_{\rm B} = 250\,\mu$m.

THE DELAY DIFFERENTIAL EQUATION AND ITS ASSOCIATED CHARACTERISTIC EQUATION

Thus far we have incorporated a total mass balance and a momentum balance into our model but we have yet to account for red blood cell transport. A coupling between red cell transport and whole blood transport has been introduced through the impact on hematocrit on hydraulic resistance. The red cell transport equations are the PDEs formulated in the previous paper by Carr and LeCoin[5]

$$\frac{\partial H_i}{\partial t} + v_i \frac{\partial H_i}{\partial x_i} = 0 \ , \tag{7}$$

where $H_i(x_i, t)$ is the hematocrit in branch i and $v_i(t)$ is the propagation velocity in branch i. This model assumes that the hematocrit is a continuous function of x_i (position along vessel i) and t. Over short time scales this is not a good assumption due to the discrete nature of RBC transport. However, over longer time scales, this is equivalent to averaging the RBC transport and smoothing out the discrete nature of the flow.

The key step in transforming the governing PDE into a DDE is to note that all of the blood in the vessel at any time was once the blood at the entrance of the vessel. We can therefore define a delay time t^* for branch i such that the blood at the vessel exit at time t was the blood at the vessel entrance at time $t - t^*$. Integrating $H_i(x_i, t)$ over the length of the vessel and dividing by the vessel length gives

$$\frac{d\bar{H}_i}{dt} = \frac{v_i}{l_i}\,(H_i(0,t) - H_i(l_i, t)) \ ,$$

$$\Rightarrow \frac{d\bar{H}_i}{dt} = \frac{v_i}{l_i}(H_i(0,t) - H_i(0, t - t^*)).$$

Applying this result to the simple network of Fig. 1 we arrive at

$$\frac{d\bar{H}_{\rm A}}{dt} = \frac{v_{\rm A}}{l_{\rm A}}\,(H_{\rm f} f(Q(t)) - H_{\rm f} f(Q(t - \tau(t)))) \tag{8}$$

$$\frac{d\bar{H}_{\rm B}}{dt} = \frac{v_{\rm B}}{l_{\rm B}}\,(H_{\rm f} g(Q(t)) - H_{\rm f} g(Q(t - \theta(t)))) \tag{9}$$

The delay times τ and θ are determined from the following integrals

$$l_{\rm A} = \int_{t-\tau(t)}^{t} v_{\rm A}(s)\, ds \tag{10}$$

$$l_{\rm B} = \int_{t-\theta(t)}^{t} v_{\rm B}(s)\, ds \tag{11}$$

Notice that in general the delay times are not constants but vary with time. This is because we anticipate that the velocity, which is directly proportional to $Q(t)$, will not be constant in time. Eqs. (8)–(11) constitute our model along with the constitutive relations discussed in the previous section.

Differential equations in which the derivative of the dependent variable depends on the instantaneous value of the dependent variable as well as its value at previous times are called delay differential equations. For a very readable introduction to delay differential equations in physiological systems see Glass and Mackey.[12] Delay differential equations in which the derivative of the dependent variable depends on a portion of the history of the dependent variable are known as state dependent delay differential equations.

In order to investigate the stability of the network, we have undertaken the following programme of study. First, we determine the steady-state solutions as discussed in the previous section. Next, we linearize the governing equations about the steady-state solution and solve the corresponding linear stability problem. In order to accomplish this, we linearize the governing PDEs, plasma skimming function, and viscosity function, which results in a set of first-order wave equations with constant (but potentially different) velocities. These can then be transformed into a set of delay differential equations with two constant delays. The full nonlinear problem including the hematocrit dependent velocity is our ultimate goal, and we use the results of the linear stability analysis to guide our search in parameter space for regions of instability.

The characteristic equation for the linear approximation is

$$1 = b\left(\frac{e^{-\lambda\gamma} - 1}{\lambda\gamma}\right) + c\left(\frac{e^{-\lambda} - 1}{\lambda}\right) , \tag{12}$$

where the parameters b, c, and γ are defined by

$$b = H_{\mathrm{f}} Q^{*}(1 - Q^{*})\frac{\beta_{\mathrm{A}}}{\mu_{\mathrm{A}}} f'(Q^{*}) , \tag{13}$$

$$c = -H_{\mathrm{f}} Q^{*}(1 - Q^{*})\frac{\beta_{\mathrm{B}}}{\mu_{\mathrm{B}}} g'(Q^{*}) , \tag{14}$$

$$\gamma = \frac{\tau}{\theta} = \frac{l_{\mathrm{A}} d_{\mathrm{A}}^{2}}{l_{\mathrm{B}} d_{\mathrm{B}}^{2}} \frac{1 - Q^{*}}{Q^{*}} . \tag{15}$$

The parameters b, c, and γ contain the information necessary to understand the linear stability of the system and we will view these parameters as functions of Q^*, parameterized by the other system parameters. Carr and LeCoin concluded that the residence times (which make up γ) were important from their simulation experiments. It would have been extremely fortuitous had they discovered b and c through dimensional analysis of simulation results. A physical interpretation of these parameters will be presented in the following section.

Our task is to find the roots λ of Eq. (12) given the values for b, c and γ. The network will be linearly unstable should there exist any root with real part of λ greater than zero. The experience of Carr and LeCoin suggest that systems can move from non-oscillating behavior to oscillations by smoothly varying lengths and/or diameters and/or inlet hematocrit. Based on that observation, we anticipate that the value of λ will pass smoothly across the imaginary axis as we vary the value of Q^* (this is equivalent to varying the length ratio). Conditions which give rise to purely imaginary λ are therefore of great interest in identifying the transition from non-oscillating systems to those that oscillate. We have been able to derive two necessary conditions for the existence of purely imaginary roots

$$-bc > 0 , \tag{16}$$

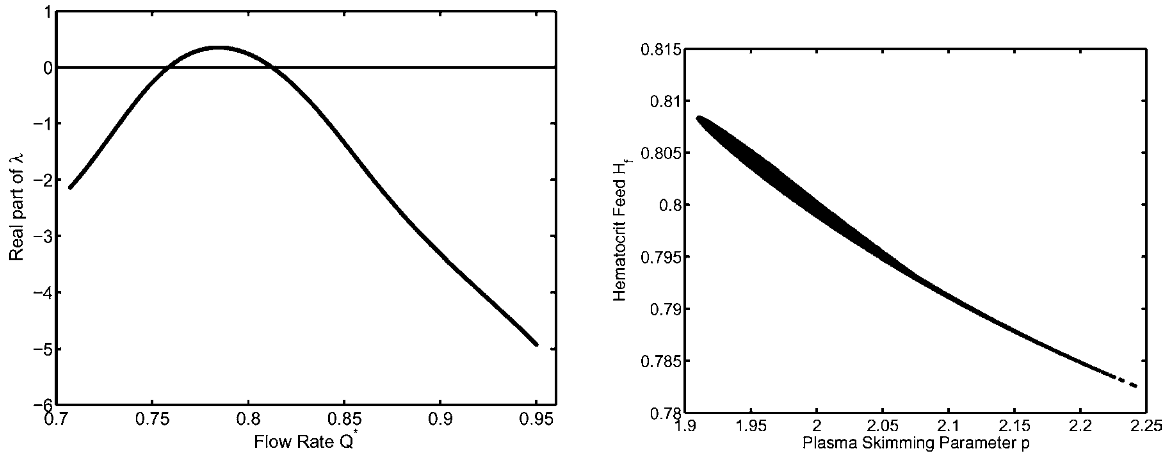
$$-bc(1 - \gamma)^2 > \gamma \, ; . \tag{17}$$

These conditions are helpful to determine if spontaneous oscillations are possible for a given set of network parameters. First, the steady-state solution(s) is computed as discussed in the previous section. Then the values of b, c and γ are calculated. If the conditions (16)–(17) are not met, no oscillations will occur. If these conditions are met, the characteristic equation (12) must be solved directly to see if the system is stable or not.

For a given set of parameters d_{A}, d_{B}, H_{f}, plasma skimming model, and viscosity model, we can plot a parametric curve of λ in the complex plane using Q^* as the parameter. In order to do this, we first search for purely imaginary roots of the characteristic equation, if they exist. We now have several points on the imaginary axis which are part of the parametric curve we seek. The rest of the points on the curve are found by iteration on the characteristic equation for complex $\lambda = \sigma + i\omega$. Any Q^* which results in λ to the right of the imaginary axis represents an unstable steady state for the network.

The following example shows how the method works to identify whether or not a network is stable. The network parameters are $H_{\mathrm{f}} = 0.8$, $d_{\mathrm{A}} = 35\,\mu\mathrm{m}$, $d_{\mathrm{B}} = 20\,\mu\mathrm{m}$, plasma skimming parameter for logit function $p = 2.0$, Pries *et al. in vitro* viscosity function. Using these network parameters, the solutions of the characteristic equation for the real part of λ using Q^* as a parameter is shown in Fig. 3a. The steady-state value for these parameters is $Q^* \approx 0.8$ which corresponds to a length ratio of ≈ 0.2. Figure 3b shows the unstable regions in a limited parameter space. In this figure the lengths and diameters are all constant while the plasma skimming parameter and the feed hematocrit are varied. Notice the very narrow region of instability.

These results have been verified by direct numerical simulation of the governing partial differential equations using the procedure of Carr and LeCoin. Figure 4a shows how the pressure at node 1 changes with time. Both vessels are initially filled with plasma but no red blood cells. In Fig. 4b we show the dynamics of the system as a phase portrait with $\bar{H}_{\mathrm{A}}$ and $\bar{H}_{\mathrm{B}}$ as parameters. In this case, both vessels are filled with blood close to steady state and the resulting dynamics is that of a limit cycle. Convergence to the non-oscillatory steady-state solution has also been confirmed when predicted by the linear stability analysis.



(a) (b)

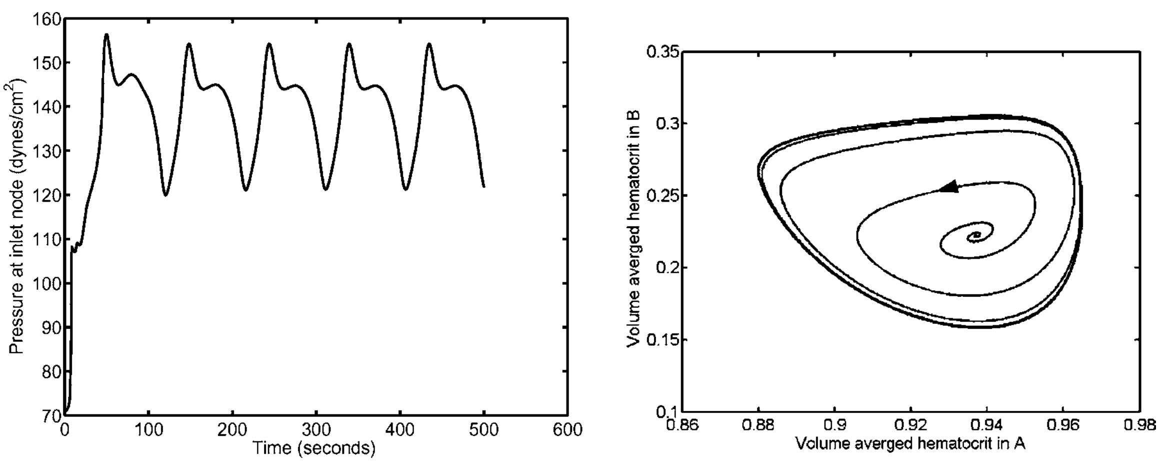
FIGURE 3. (a) The real part of λ versus Q^* with the following parameters: $H_f = 0.8$, plasma skimming parameter $p = 2.0$, *in vitro* viscosity model, $d_A = 35\,\mu$m, and $d_B = 20\,\mu$m. Steady state corresponds to $Q^* \approx 0.8$ which corresponds to a length ratio of approximately 0.2. (b) The region of instability in the $p - H_f$ plane with $H_f = 0.8$, plasma skimming parameter $p = 2.0$, *in vitro* viscosity model, $d_A = 35\,\mu$m, $d_B = 20\,\mu$m, $l_A = 50\,\mu$m, and $l_B = 250\,\mu$m. Notice that the region of instability is quite narrow.

DISCUSSION AND INTERPRETATION OF RESULTS

The network shown in Fig. 1 is the simplest network that can exhibit spontaneous oscillations. Nevertheless, there are still five physical variables: p (plasma skimming parameter); two branch diameters (d_A and d_B); the feed hematocrit to the network (H_f); the ratio of branch lengths (l_A/l_B). In addition, a viscosity function is required. Both diameters are needed rather than just the ratio of the diameters because the Fahraeus–Lindqvist effect requires information about each branch diameter. Normalizing and linearizing the governing equations reduces this set of parameters to three dimensionless groups which we have named b, c and γ. For complex λ with real part greater or equal to 0, it is necessary for b and c to have opposite signs,

$$-bc > 0\,.$$

The defining equation for b shows that the sign of b depends on the slope of the plasma skimming function f' since all other terms in the equation are positive. The same is true for c. Its sign is determined by $-g'$. For b and c to have opposite signs, f' and g' must therefore both have the same sign. Figure 5 shows a plot of $f(Q^*)$ and $g(Q^*)$. From



(a) (b)

FIGURE 4. Simulation results for network with $H_f = 0.8$, plasma skimming parameter $p = 2.0$, *in vitro* viscosity model, $d_A = 35\,\mu$m, $d_B = 20\,\mu$m, $l_A = 50\,\mu$m, $l_B = 250\,\mu$m. The corresponding steady-state flow rate is $Q^* \approx 0.8$. (a) Pressure fluctuations at the inlet node as a function of time. (b) Phase portrait showing a limit cycle. Volume averaged hematocrit in B is plotted against volume averaged hematocrit in branch A.

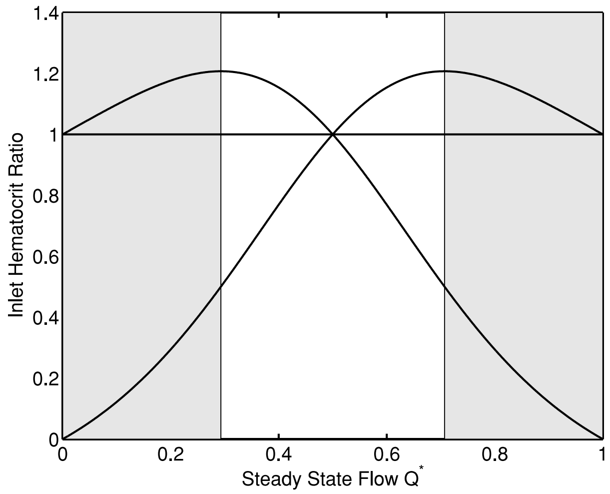


FIGURE 5. Example of plasma skimming rule for branches A and B with plasma skimming parameter $p = 2$. No oscillations can occur in the unshaded region.

$Q^* = 0$ to $Q^* = Q_1$, the slopes of both $f(Q^*)$ and $g(Q^*)$ are both positive. From $Q^* = Q_1$ to $Q^* = Q_2$, the slope of $f(Q^*)$ is positive and the slope of $g(Q^*)$ is negative. From $Q^* = Q_2$ to $Q^* = 1$, the slopes of both curves are negative. Notice that Q_1 is at the maximum for $g(Q^*)$ and Q_2 is the maximum of $f(Q^*)$. This graph shows that over the interval of $Q_1 < Q^* < Q_2$ no oscillations are possible. We have a physical interpretation for this conclusion.

Consider that the steady-state fractional flow is between Q_1 and Q_2. Imagine that the system is at steady state and the fractional flow is perturbed (total inlet flow remains constant). A positive perturbation in Q^* results in an increase in $\bar{H}_A$ and a decrease in $\bar{H}_B$. This means the ratio R_A/R_B increases and the system responds to the perturbation by moving back toward the steady state. A negative perturbation in Q^* results in a decrease in $\bar{H}_A$ (and R_B) and an increase in H_B (and R_B). Again the system responds to the perturbation by moving back toward the steady state. If the b and c for a given steady state have the same signs, the steady state is necessarily stable. If $Q^* < Q_1$, then a positive perturbation in Q^* results in increases in both $\bar{H}_A$ and $\bar{H}_B$ (as well as in R_A and R_B). If $Q^* > Q_2$, then a positive perturbation leads to decreases in $\bar{H}_A$, R_A, $\bar{H}_B$ and R_B. This may or may not lead to an instability. The answer depends in part on how the ratio R_A/R_B responds to the perturbation. It turns out that the fractional change in R_A in response to a perturbation in Q^* is related to b, while the fractional change in R_B is related to c via

$$\frac{\Delta R_A}{R_A} = \frac{\beta_A}{\mu_A}\frac{f'(Q^*)}{\tau} = \frac{b}{H_f}Q^*(1-Q^*)\ , \tag{18}$$

$$\frac{\Delta R_B}{R_B} = \frac{\beta_B}{\mu_B}\frac{g'(Q^*)}{\theta} = \frac{-c}{H_f}Q^*(1-Q^*)\ , \tag{19}$$

which clearly demonstrates that the parameters b and c are imbued with significant physical meaning.

The analysis of the linearized delay differential equation provided another necessary condition for oscillations,

$$-bc(1-\gamma)^2 > \gamma\ . \tag{20}$$

We have already seen some of the physical significance of the $-bc$ term and the γ term. The $1-\gamma$ term also has some physical significance. This term is related to the difference in steady-state delay times. When the delay times are equal, no oscillations are possible, regardless of the values of b and c, since any perturbation will damp out. The mechanism for this is as follows. The perturbation from steady state will cause changes in the hematocrit of the blood entering the branches. A slug of blood entering the diverging node will split into two segments in the daughter branches. If the delay times (residence times) are equal, those same two segments will recombine with each other at the exit of the loop in the converging node. This means that the hematocrit in the exit branch will have uniform and constant hematocrit because the rate at which red cells are loaded into the loop equals the rate at which the red cells are unloaded from the loop. This has the effect of stabilizing the system. Now, if the delay times are not equal, the two segments formed at the entrance of the loop recombine with other segments at the exit, and this can lead to sustained oscillations. Finally, we note that this second condition, $-bc(1-\gamma)^2 > \gamma$, must be satisfied for the existence of purely imaginary eigenvalues. Although we have not proven it, we have observed that this condition is satisfied for all cases we have found with complex λ with positive real part.

Our analysis is clearly incomplete. For example, none of conditions on b, c and γ are sufficient. In addition, we have not been able to show that a combination of these necessary conditions is sufficient to predict oscillations. Therefore we use the necessary conditions to screen out situations for which sustained oscillations are not possible. If all of these necessary conditions are met, we still need to use the characteristic equation to determine whether the system has sustained oscillations or not.

Our theory is not dependent on any particular model for the viscosity function. We have found oscillating systems using the *in vivo* and *in vitro* models of Pries *et al.*[21] and a viscosity function which is linear in hematocrit. The development of our method is independent of any particular plasma skimming model as well. We note, however, that the linear plasma skimming model used by Carr and LeCoin[5] as well as by Kiani *et al.*[15] cannot result in oscillations with our topology. For the linear plasma skimming model the product bc is always positive or zero, never negative. It is true the linear plasma skimming model can produce oscillations in slightly more complicated topologies. The addition of one more entrance branch to our topology is enough for the straight line plasma skimming model to result in instabilities.

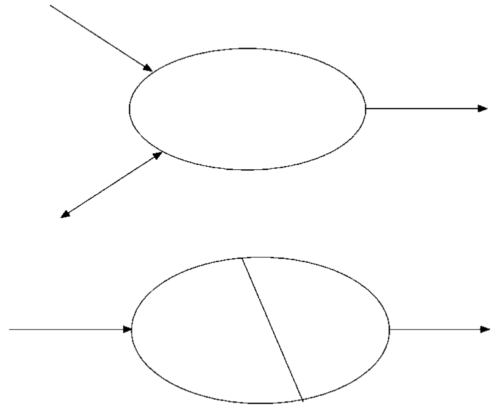


FIGURE 6. Other network topologies that can be analyzed using the techniques of this paper. Numerical simulations suggest that the topology shown in the top panel can have oscillations with the straight line plasma skimming model. The bottom panel has nodes which are not connected to any external nodes.

Although the topology considered in this study is very simple, the methods and ideas can be directly applied to more complicated networks such as those shown in Fig. 6. It is unlikely, however, that these methods could be applied to very large, physiologically realistic networks. On the other hand, by analyzing the minimal network topology that allows for spontaneous oscillations we have been able to identify some of the key ingredients that are likely to lead to nonlinear oscillations in such networks.

There are several difficulties in experimentally confirming the results of this analytical and numerical study. One of the problems is that the parameters b, c, and γ are not independent. The computation of b and γ require all information about the network (lengths, diameters, feed hematocrit, and plasma skimming parameter). The evaluation of c requires all information about the network except the diameter of branch A. This makes independent tuning of the physical parameters in a network to satisfy the conditions needed for oscillations an impossible task. We have used the theory outlined in this paper to guide our search through parameter space to find oscillating networks.

A second difficulty in experimentally confirming this study is the extremely narrow range of feed hematocrit over which oscillations occur. We are currently unable to control inlet hematocrit to this degree of precision. However, preliminary analysis suggests that slightly more complex topologies may demonstrate oscillations over a wider range of parameter space making experimental confirmation possible.

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